

NATIONAL SCIENCE FOUNDATION SBIR PHASE I FULL PROPOSAL DRAFT

SINE~WaiV: Software-Only Motor Current Signature Analysis for VFD-Driven Industrial Motors

AI BASE3 LLC | SINE~WaiV | Canonical Synthesis Draft

Solicitation	NSF 26-510 - Small Business Innovation Research / Small Business Technology Transfer Phase I, Phase II, Fast-Track Programs: Developing Deep Technologies that Advance U.S. Competitiveness and Security
Deadline Target	July 27, 2026, 5:00 PM submitting organization's local time
Project Title	SINE~WaiV: Software-Only Motor Current Signature Analysis for VFD-Driven Industrial Motors
Proposing Firm	AI BASE3 LLC, Lemay, St. Louis, Missouri 63125
Principal Investigator	Anthony McGee, Founder and Solutions Engineer
Project Pitch Reference	#00113162 - invited under Advanced Systems for Scalable Analytics
Requested Amount	\$305,000
Proposed Duration	12 months
Company Identifiers	UEI: K8E5DCCSS833 EIN: 42-2737209 Missouri Charter: LC014726944
Draft Status	Canonical synthesis draft for founder, advisor, and reviewer feedback. Not yet a Research.gov submission package.

Project Summary

Overview. AI BASE3 LLC proposes to establish the technical feasibility of SINE~WaiV, a software-only Motor Current Signature Analysis (MCSA) engine for variable-frequency-drive (VFD) driven industrial motors. Modern VFDs improve energy efficiency but introduce pulse-width-modulated carrier artifacts that can bury weak current-domain indicators of bearing, rotor, and load-related degradation. Existing high-confidence predictive maintenance solutions often avoid this difficulty by adding vibration sensors, proprietary acquisition hardware, installation labor, and recurring diagnostic infrastructure. SINE~WaiV instead evaluates existing motor current, $i(t)$, through a constrained pipeline: $i(t) \rightarrow \hat{X}(f,t) \rightarrow S(t) \rightarrow RUL(t)$. The Phase I project will test whether this software layer can improve fault-feature observability without adding sensors or building custom deployment hardware.

Intellectual Merit. The proposed architecture combines a 65,536-point primary FFT at 20 kHz sampling, producing approximately 0.305 Hz/bin resolution, with a 1,728-point mixed-radix sub-window layer factored as $2^6 \times 3^3$. The 1,728-point structure provides 28 valid integer divisors that allow deterministic sub-window steering around targeted VFD artifact regions, including 170-185 Hz and 258-260 Hz. The research does not claim that FFTs, mixed-radix computation, or MCSA are new. The contribution is the controlled use of this architecture for VFD carrier-artifact avoidance and current-domain fault-feature extraction. NASA IMS will be used only as an accelerometer-domain temporal proxy benchmark. Current-domain validation will be performed on Paderborn motor-current data and synthetic VFD artifact injection tests. Independent review will evaluate signal-to-noise improvement, detection latency, false-positive behavior, and computational feasibility.

Broader Impacts. If successful, SINE~WaiV would reduce the cost and complexity of industrial predictive maintenance by allowing smaller manufacturers, municipal utilities, and lower-margin operators to extract earlier warning from electrical data already present in motor-control infrastructure. The commercialization path is an OEM-agnostic software overlay for field-bus-accessible VFD and motor-control systems, followed by Phase II field validation, reliability-service channels, and licensing to automation platforms.

1. Identification and Significance of the Innovation

Industrial predictive maintenance has a deployment problem. The physical failure information needed to diagnose motors often exists before breakdown, but extracting that information across a real plant is expensive, inconsistent, and highly dependent on specialized instrumentation. Vibration monitoring has become the dominant practical method because it measures mechanical motion directly and has a long history of validated bearing and rotor-fault relationships. However, vibration monitoring requires physical access, mounted sensors, wiring, installation labor, ongoing calibration, and, in many cases, a recurring cloud platform. These costs are manageable for large enterprise facilities with reliability engineering departments. They are not manageable for many small and mid-size manufacturers, water plants, food processors, aggregate operations, university infrastructure departments, and other industrial operators with hundreds of motors but limited diagnostic staff.

MCSA offers a more scalable path because the motor itself functions as a transducer. Rotor eccentricity, bearing defects, mechanical imbalance, load modulation, and air-gap disturbances alter electromagnetic behavior, and those effects appear as sidebands or other modulated structures in stator current. In principle, this allows condition monitoring from current transformers and VFD data already present in motor-control infrastructure. In practice, VFD-driven systems make the problem substantially harder. PWM switching produces drive-induced spectral components and broadband distortion that can mask or mimic weak mechanical fault features. Conventional fixed-window FFT analysis can either miss the fault signature, smear the signature across bins, or treat VFD artifact energy as diagnostic structure. Filtering can reduce the artifact but also removes information that may be diagnostic.

The significance of SINE~WaiV is that it attacks the cost and observability problem at the software layer. The target is not a new sensor package. The target is a validated algorithmic method that can improve current-domain feature extraction under the spectral conditions created by modern drives. A successful Phase I result would support a credible transition from high-cost hardware deployments toward software-assisted, field-bus-accessible diagnostic coverage. That transition matters because VFDs are not a niche installation; they are a core mechanism for energy-efficient industrial speed control in pumps, fans, conveyors, compressors, and process equipment.

The innovation is deliberately narrow. SINE~WaiV does not claim that a mixed-radix FFT is new, that MCSA is new, or that VFDs have never been analyzed. The claim is that a constrained 1,728-point mixed-radix sub-window layer, operating on a high-resolution spectral canvas and steered around specific carrier-artifact bands, can improve the extraction of current-domain fault structure in a way that is computationally suitable for edge deployment. The method is designed to preserve physically interpretable sideband information rather than depend on black-box anomaly scoring. Its value will be measured by whether it produces superior current-domain observability under VFD contamination, not by mathematical novelty of the FFT itself.

The preliminary internal result motivating this proposal is that the same structural scoring approach produced an approximately 134-hour warning horizon on the NASA IMS run-to-failure benchmark when used as a degradation-timing proxy, substantially earlier than a simple threshold approach in that internal comparison. Because NASA IMS is accelerometer-domain data, that result cannot be treated as proof of current-domain performance. It is used here only to justify the Phase I research question: if the temporal coherence behavior can be tracked in a well-known run-to-failure benchmark, can the architecture also isolate motor-current fault indicators under realistic VFD artifact conditions? Phase I is designed to answer that question directly.

2. Technical Background and State of the Art

Three bodies of work support the proposed research. The first is classical FFT-based spectral analysis. Cooley-Tukey FFT decomposition and later mixed-radix implementations are mature, well-understood, and widely implemented. Their maturity is an advantage for this proposal because the computational behavior can be inspected, profiled, and validated without inventing a new transform. The novelty is in task-specific configuration and deployment: using a prime-factor-compatible window structure to give the

diagnostic engine more valid integer resolution choices in the specific bands where VFD artifacts interfere with current-domain fault features.

The second body of work is MCSA and ESA for induction motors. Prior literature has shown that motor current can contain signatures associated with bearing defects, rotor bar damage, eccentricity, load variation, and other mechanical or electrical faults. ISO 20958 provides a formal standards context for electrical signature analysis of three-phase induction motors. This is important because SINE~WaiV is not attempting to replace the field of ESA/MCSA; it is attempting to address a specific barrier that limits broader deployment in VFD-driven environments. The proposal therefore positions traditional ESA/MCSA as the technical foundation, not as an adversary.

The third body of work is condition monitoring under non-ideal drive and load conditions. IEEE 519-2022 provides a current reference point for power-system distortion at the point of common coupling, but predictive diagnostics require more than steady-state power-quality compliance. A motor may operate within acceptable distortion limits and still contain weak fault information that is difficult to separate from drive-induced spectral components. Similarly, vibration monitoring may remain robust where MCSA struggles, but vibration robustness is purchased through added hardware and installation labor. The market gap exists precisely because neither conventional MCSA nor hardware-heavy vibration monitoring provides a low-friction, software-only solution that scales across the installed base.

Market leaders and incumbent platforms are therefore best understood as evidence of need rather than evidence against the opportunity. Conventional vibration systems offer high diagnostic confidence but require sensors and field installation. Established ESA/MCSA systems offer a pathway through electrical measurements but often rely on proprietary acquisition hardware, expert interpretation, or operating conditions that do not generalize cleanly across VFD environments. Emerging machine-learning platforms can detect anomalies, but they may require substantial historical failure data, may struggle with explainability, and may not distinguish a physically meaningful fault sideband from a drive artifact unless the feature extraction layer is already disciplined.

SINE~WaiV's contribution is a deterministic observability layer. The engine does not begin by asking a model to infer a failure class from arbitrary plant data. It begins by structuring the spectrum so that known mechanical fault relationships have a better chance of being visible. The Phase I effort will compare this approach against fixed-window FFT baselines and simple threshold methods. The result may support, modify, or reject the central hypothesis. That falsifiability is essential: if current-domain features cannot be separated with sufficient margin under the proposed artifact conditions, the Phase I research will document the boundary and prevent premature commercialization.

3. SINE~WaiV Technical Architecture

SINE~WaiV operates on motor current, $i(t)$, with the goal of producing interpretable state information from existing electrical measurements. The core pipeline is: $i(t) \rightarrow \hat{X}(f,t) \rightarrow S(t) \rightarrow RUL(t)$. The input signal is a time-domain motor current waveform. The spectral estimator $\hat{X}(f,t)$ produces time-indexed frequency-domain structure. The state score $S(t)$ compresses selected spectral features into a degradation or fault-state indicator. The $RUL(t)$ output is not a universal prediction of exact failure time; it is a bounded warning horizon or remaining-useful-life estimate derived from the evolution of $S(t)$ under documented operating assumptions.

The first stage is a 65,536-point FFT sampled at 20 kHz, producing approximately 0.305 Hz/bin resolution. This stage establishes a high-density spectral canvas. The point is not simply to make the transform larger. The point is to create enough spectral granularity that weak sidebands near drive-induced artifacts are not immediately lost to coarse bins. The Phase I implementation will define memory allocation, leakage control, window selection, and repeatability constraints so that the same waveform produces the same feature estimates under controlled conditions.

The second stage is a 1,728-point mixed-radix sub-window framework factored as $2^6 \times 3^3$. This configuration yields 28 valid integer sub-window divisors. A binary-only 2^N structure provides efficient computation but limits divisor flexibility. The 1,728-point layer provides a constrained set of window

choices that can be scheduled away from known artifact regions while preserving enough resolution to track sideband behavior. The targeted VFD carrier-artifact regions for Phase I are 170-185 Hz and 258-260 Hz. These are treated as empirical suppression bands, not universal constants. The algorithmic task is to demonstrate that the sub-window scheduler can reduce artifact interference in these bands without erasing the fault-modulated information adjacent to them.

The third stage is feature extraction and state scoring. Rather than relying on a single amplitude threshold, SINE~WaiV will evaluate sideband stability, local spectral contrast, feature persistence across adjacent windows, and temporal coherence decay. The scoring layer will be designed to separate persistent structural change from transient operating changes. This is where field realism matters: production loads change, line conditions change, and drives can generate nonstationary artifacts. The Phase I scoring layer will therefore log the conditions under which a score is valid, uncertain, or rejected.

The fourth stage is RUL or warning-horizon estimation. The draft deliberately avoids an overbroad claim that SINE~WaiV can predict exact failure time for every motor from a single scan. Phase I will instead ask whether the system can increase actionable warning time relative to fixed-window FFT baselines under defined datasets and contamination regimes. A valid Phase I success does not require perfect failure-time prediction. It requires measurable improvement in observability, detection latency, and false-positive control while remaining computationally practical.

The Phase I scope is software research and validation. Bench-level hardware validation, custom edge gateways, and deployment hardware are reserved for Phase II activities pending successful Phase I results. This prevents scope drift and makes the proposal testable inside the Phase I budget and timeline.

4. Research Hypotheses, Success Criteria, and Failure Criteria

Hypothesis	Primary Metric	Success Threshold	Failure Boundary
H1 Artifact Suppression	Local SNR in 170-185 Hz and 258-260 Hz	At least 8 dB median gain; 12-16 dB stretch	Below 3 dB or loss of adjacent fault information
H2 Current Transfer	Paderborn current-channel feature visibility	Repeatable current-domain sideband extraction	NASA-only support or no current-domain separation
H3 Warning Horizon	Detection latency against baselines	Earlier persistent detection than fixed-window FFT	Retrospective tuning only or unstable warnings
H4 Computational Fit	Memory, runtime, reproducibility	Bounded, repeatable execution under resource envelope	Unbounded memory/runtime or non-reproducible outputs

The proposal is organized around four falsifiable hypotheses.

H1 - Artifact Suppression Hypothesis. When evaluated on current-domain motor data contaminated by VFD-like carrier artifacts, the SINE~WaiV mixed-radix sub-window scheduler will improve local signal-to-noise ratio in the targeted 170-185 Hz and 258-260 Hz bands compared with a fixed-window FFT baseline. The primary success threshold is a median improvement of at least 8 dB across the targeted test cases, with a stretch target of 12-16 dB. A result below 3 dB median improvement, or a result that improves one band only by destroying adjacent fault information, will be considered a technical failure for this hypothesis.

H2 - Current-Domain Transfer Hypothesis. Degradation-sensitive features identified through the SINE~WaiV pipeline will be extractable from motor-current data in the Paderborn bearing dataset after appropriate channel selection, decimation, and metadata alignment. NASA IMS is not used to prove current-domain performance. NASA IMS is used only as a temporal proxy benchmark for degradation timing. Current-domain validation must stand on Paderborn current-channel analysis and on synthetic VFD artifact injection into current-domain records. Failure to show current-domain separation will prevent any Phase II claim of sensor-free deployment.

H3 - Detection Latency Hypothesis. SINE~WaiV will detect a persistent degradation indicator earlier than fixed-window FFT or simple threshold baselines under the same data and scoring conditions. The primary success metric is warning lead time measured as the elapsed time between first persistent detection and failure or labeled defect progression. The result will be reported with confidence intervals where the

dataset supports them. A warning that appears only after heavy retrospective tuning will not count as success.

H4 - Computational Feasibility Hypothesis. The engine will execute with bounded memory use and repeatable runtime behavior suitable for later edge implementation. Phase I does not require deployment on custom hardware. It does require profiling under a simulated edge-resource envelope and documenting whether the core routine can operate within realistic industrial data rates. Success requires stable memory behavior, reproducible outputs, and a runtime profile that does not require cloud-scale processing for each motor.

The success criteria are intentionally conservative. The project will not be judged by a single attractive chart. It will be judged by reproducibility, data-domain discipline, and clearly documented limitations. A strong Phase I outcome is one in which the engine shows measurable artifact avoidance, current-domain fault visibility, and bounded computational cost. A useful negative outcome is also possible: the research may show that the architecture works as a temporal degradation tracker but does not transfer adequately to current-domain VFD data. In that case, the Phase I result would still define a scientifically useful boundary and prevent wasteful commercialization claims.

5. Phase I Work Plan and Technical Tasks

Task	Timing	Core Activity	Deliverable
1	Months 1-3	Canonical build and baseline reproduction	Locked code, parameter registry, baseline protocol
2	Months 3-5	NASA IMS temporal proxy benchmark	Timing benchmark and boundary statement
3	Months 5-8	Paderborn current-domain validation	Current-channel SNR and sideband report
4	Months 7-10	Synthetic VFD artifact injection	Pass/fail artifact suppression report
5	Months 9-11	Independent review and falsification	External validation memorandum
6	Months 10-12	Commercialization and Phase II readiness	Customer discovery log and Phase II plan

The proposed Phase I period is 12 months. The work plan is sequenced so that implementation, benchmark validation, current-domain validation, and commercialization evidence do not collapse into one uncontrolled effort.

Task 1: Canonical Software Build and Baseline Reproduction, Months 1-3. The PI will build the canonical SINE~WaiV research implementation in a controlled repository. The software will include the 65,536-point primary FFT, the 1,728-point mixed-radix sub-window scheduler, signal pre-processing controls, feature extraction routines, state-scoring routines, and structured logging. The same period will establish fixed-window FFT baselines, simple threshold baselines, leakage-control procedures, and reproducible test notebooks. Deliverables include a documented research build, a locked parameter registry, a baseline comparison protocol, and an internal reproducibility report.

Task 2: Temporal Proxy Benchmarking on NASA IMS, Months 3-5. NASA IMS will be analyzed only as an accelerometer-domain run-to-failure timing benchmark. The purpose is to verify whether the scoring layer can track degradation timing and warning-horizon behavior against a recognized degradation sequence. The task will not be used to claim that vibration-domain performance proves MCSA performance. Deliverables include a timing benchmark report, a comparison with threshold baselines, plots of $S(t)$ over degradation time, and a documented explanation of what NASA IMS does and does not validate.

Task 3: Current-Domain Validation on Paderborn, Months 5-8. The Paderborn University dataset contains motor-current and vibration channels under controlled electromechanical conditions. Phase I will focus on the current channel, with decimation and metadata handling documented in the methods. The objective is to test whether SINE~WaiV can extract fault-modulated current-domain features under known bearing conditions and compare those features against fixed-window FFT results. Deliverables include current-

channel preprocessing scripts, sideband mapping tables, local SNR calculations, false-positive review, and a current-domain validation report.

Task 4: Synthetic VFD Artifact Injection, Months 7-10. Because public current-domain datasets may not fully reproduce the targeted VFD artifact conditions, the project will conduct synthetic injection tests on current-domain data. The injection protocol will create repeatable artifact patterns in and around the 170-185 Hz and 258-260 Hz bands, at multiple amplitudes and noise conditions. The purpose is not to fabricate performance but to stress-test the scheduler against controlled contamination. Deliverables include an injection protocol, artifact parameter table, before-and-after spectral comparisons, and pass/fail results for H1.

Task 5: Independent Technical Review and Falsification, Months 9-11. AI BASE3 will retain an external technical reviewer or consultant with signal-processing, electrical machines, or condition-monitoring expertise. The reviewer will receive the locked methods, datasets or dataset access instructions, and compiled outputs sufficient to evaluate claims without relying on founder interpretation. The reviewer will calculate or verify SNR improvement, detection latency, and false-positive behavior. Deliverables include an independent validation memorandum, a list of claims supported by data, and a list of claims that must be removed or delayed until Phase II.

Task 6: Commercialization and Phase II Readiness, Months 10-12. The final task will translate the technical findings into a Phase II readiness package. The PI will conduct structured customer discovery with industrial operators, maintenance leaders, automation vendors, and reliability service providers. The goal is to test buying behavior, integration friction, and licensing assumptions. Deliverables include a customer discovery log, a Phase II field-trial design, a revised product requirements document, and a final NSF Phase I technical report.

6. Experimental Design and Data Management

The experimental design separates three data categories to prevent domain confusion.

Category A is temporal proxy data. NASA IMS belongs in this category. It is valuable because it is a recognized run-to-failure benchmark, but it is accelerometer-domain data. Results from Category A can support statements about degradation timing behavior, scoring stability, and warning-horizon modeling. They cannot support a claim that SINE~WaiV has proven current-domain VFD performance.

Category B is current-domain validation data. Paderborn belongs in this category. The current-channel records will be processed through the same pipeline used for Category A, but the performance claims will be stated separately. This category is where MCSA feasibility begins to be tested. The key outputs are sideband visibility, spectral separation, and repeatability of feature extraction under known bearing conditions.

Category C is controlled contamination data. Synthetic VFD artifact injection belongs in this category. It allows the project to test how the scheduler behaves when artifact energy is deliberately placed in the kill zones. Because the injection parameters are controlled, the project can quantify whether improvement is due to real separation or simply to suppression that removes both artifact and fault information.

For all categories, the project will preserve raw input references, preprocessing scripts, parameter settings, feature outputs, and derived score files. Where the data source is public, the document will cite access instructions rather than redistributing proprietary copies. Where field or partner data is later obtained, it will be de-identified and stored with metadata fields for motor rating, drive type, sampling rate, load condition, and acquisition context. File formats will include CSV, HDF5, JSON metadata, and versioned analysis notebooks.

Data generated under the award will be managed consistent with NSF SBIR data rights. Public benchmark results, high-level validation statistics, and non-proprietary methods summaries can be shared in final reports and reviewer materials. Proprietary source code, optimized scheduler logic, feature selection heuristics, trained scoring parameters, and commercial product integration details will remain AI

BASE3 trade secret material unless later protected through patent filings or disclosed under controlled partner agreements.

7. Expected Outcomes and Technical Risk Management

The expected Phase I outcome is a defensible technical feasibility decision. The strongest positive outcome would show that SINE~WaiV produces consistent current-domain SNR improvement in targeted VFD artifact bands, detects persistent degradation indicators earlier than baseline approaches, and operates within a computational envelope suitable for later deployment inside existing industrial infrastructure. That outcome would justify Phase II field trials and partner integration.

The project also identifies clear failure modes. First, the sub-window scheduler may reduce artifact energy but also remove adjacent fault information. In that case the method would not be commercially useful. Second, NASA IMS timing behavior may not transfer into motor-current data. That would require narrowing the commercial claim to timing analytics or returning to feature-engineering research. Third, Paderborn current-channel results may show fault visibility only under clean laboratory conditions but not under injected VFD contamination. That would push the work toward Phase II hardware and field data collection rather than immediate software licensing. Fourth, the computational profile may be too heavy for edge deployment without substantial optimization. That would require a revised implementation strategy.

Risk mitigation is built into the task sequence. NASA IMS is isolated to prevent overclaiming. Paderborn is used for current-domain validation. Synthetic injection creates controlled stress conditions. Independent review reduces founder confirmation bias. Conservative success thresholds prevent a single cherry-picked result from being treated as proof. Commercialization work is delayed until the technical boundary is clearer but not ignored, because NSF expects a path to product and market.

This disciplined risk posture is especially important because the PI is a solo founder. The proposal directly addresses that risk by limiting Phase I to software research, reproducibility, independent validation, and customer discovery. It does not require construction of a custom test stand, a new sensor platform, or a full commercial product. It asks for enough funding to answer a high-value technical question that cannot be answered reliably with unfunded effort alone.

8. Company, Team, and Execution Capability

AI BASE3 LLC is an early-stage Missouri company formed to commercialize SINE~WaiV and related industrial signal intelligence technologies. The company is led by Anthony McGee, Founder and Solutions Engineer, who brings more than 27,000 hours of industrial electrical maintenance, diagnostic troubleshooting, automation, and plant-floor systems experience. This experience is directly relevant because the proposed technology is not an abstract analytics tool for clean laboratory data. It is intended for the electrical and mechanical noise, incomplete documentation, cost pressure, and downtime risk that define real industrial environments.

The PI's practitioner background is a strategic asset. A large fraction of predictive maintenance failure occurs not in the algorithm but in deployment assumptions: inaccessible machines, inconsistent wiring, variable loads, plant-floor contamination, limited technical staffing, and capital constraints. The PI has direct experience with those constraints. That allows the research plan to focus on what a plant can actually adopt: software that uses existing motor-control infrastructure, produces interpretable outputs, and avoids unnecessary hardware installation.

The execution risk is that a solo founder may lack academic depth or independent validation capacity. The proposal addresses this risk directly. The PI will retain control of core algorithm architecture and implementation, while an external technical consultant will review methods, run or verify calculations, and identify unsupported claims. Commercialization and customer-discovery work will be supported through regional small-business and manufacturing-support networks as appropriate. This creates a lean structure that directs Phase I funding toward R&D and validation rather than administrative overhead.

At the time of award, the PI will meet NSF primary-employment requirements for the small business. The proposal is an SBIR, not an STTR, and therefore does not require a research-institution subaward. Academic or technical advisors may be retained as consultants for independent validation, but the core commercialization pathway remains with AI BASE3 LLC.

9. Commercialization Strategy

The commercial thesis is that predictive maintenance becomes more scalable when it stops depending on new physical sensor deployment for every asset. SINE~WaiV is designed as a software layer that can eventually operate on field-bus-accessible VFD and motor-control data, or as a low-footprint runtime connected to existing current measurement infrastructure. This model allows AI BASE3 to sell value as improved observability and reduced downtime rather than as another hardware installation project.

The first market path is brownfield industrial deployment. Many plants already operate VFD-driven pumps, fans, compressors, conveyors, and process assets. They may also have motor-control centers, current transformers, VFD logs, and industrial PCs, but they may not have high-density vibration monitoring across every motor. A software-only or software-primary diagnostic layer can be piloted on a subset of critical motors, validated against maintenance records, and expanded without the same installation friction as sensor networks.

The second market path is OEM or platform licensing. VFD manufacturers, motor-control vendors, automation software companies, and reliability platforms all have incentives to improve built-in diagnostics. SINE~WaiV can be offered as a spectral feature extraction and state-scoring engine that strengthens existing platforms rather than forcing immediate replacement. This matters because established plants are reluctant to rip out platforms they already operate. SINE~WaiV is therefore positioned as a lightweight, OEM-agnostic software overlay that can be deployed on existing field-bus-accessible VFDs without requiring replacement of installed condition-monitoring systems, thereby lowering integration switching costs for plants already using incumbent reliability platforms.

The third market path is reliability-service enablement. Industrial service providers can use the engine to improve diagnostic coverage for customers that cannot justify enterprise sensor deployments. This creates a channel in which AI BASE3 licenses software or analytics capacity while service providers handle customer relationships, on-site context, and maintenance workflows.

The initial revenue model is expected to combine pilot fees, per-motor or per-site subscriptions, and OEM licensing. Phase I will not attempt to finalize pricing. It will test willingness to evaluate the technology, identify the minimum proof required for a paid pilot, and document the integration objections raised by maintenance managers and automation vendors. Phase II readiness will depend on technical results plus evidence that customers will participate in field trials.

The defensibility of the business is built from four layers. First, the implementation-specific scheduler, artifact maps, feature selection logic, and scoring parameters can be protected as trade secrets. Second, the company can pursue patent protection around the constrained sub-windowing method for VFD carrier-artifact suppression if prior-art review supports filing. Third, the product will accumulate a proprietary database of de-identified spectral features and maintenance outcomes over time. Fourth, the founder's domain-specific understanding of plant-floor deployment gives the company a practical advantage over generic analytics tools that lack industrial context.

10. Broader Impacts

The broader impact of SINE~WaiV is the democratization of predictive maintenance. Advanced condition monitoring is not evenly distributed across industry. Large enterprises can pay for vibration networks, cloud analytics, and reliability specialists. Smaller operators often rely on preventive replacement intervals, experienced technicians, audible or thermal signs of trouble, or reactive repair after failure. This gap matters because unplanned motor failures can stop production, damage downstream equipment, waste energy, and create safety risk.

A validated software-first MCSA layer can lower the cost of entry. If the technology succeeds, facilities with existing motor-control infrastructure could gain earlier warning without installing a new sensor on every motor. Municipal utilities could monitor pumps that are difficult or expensive to instrument. Smaller manufacturers could allocate maintenance labor based on condition rather than calendar intervals. University and public infrastructure departments could improve reliability without enterprise-scale reliability budgets.

The workforce impact is also important. SINE~WaiV is not meant to replace electricians, mechanics, or reliability technicians. It is meant to give them a better signal. A clear diagnostic score, supporting spectral evidence, and a recommended action window can help non-specialist plant personnel make better decisions without requiring every facility to employ a signal-processing engineer. In that sense, the tool acts as a workforce multiplier.

Environmental benefits may follow from earlier detection and better maintenance scheduling. Motors that degrade mechanically often draw energy less efficiently, generate heat, increase vibration, and shorten the life of adjacent components. Avoiding catastrophic failures can reduce scrap, emergency shipping, unnecessary replacement, and downtime energy waste. The Phase I proposal will not quantify these impacts as proven outcomes. It will establish the technical basis required before such impacts can be credibly measured in Phase II field trials.

11. Budget Justification Summary

Budget Area	Purpose	Phase I Boundary
PI personnel	Founder-led algorithm build, validation, reporting	Meets or exceeds NSF PI effort requirement
Technical support	Implementation, data handling, reproducibility	Software and analysis only
Independent consultant	External validation of SNR, latency, and methods	Reduces solo-founder confirmation risk
Computer services	Compute, storage, versioning, analysis notebooks	No custom deployment hardware
Travel/customer discovery	Industrial interviews and Phase II pilot definition	Not general sales activity
Indirect/fee	Allowable administration and small business fee	To be finalized in Research.gov budget

The requested Phase I budget is \$305,000 for a 12-month feasibility project. The budget is structured to fund software research, reproducible validation, independent technical review, customer discovery, and preparation for Phase II field evaluation. It intentionally excludes custom hardware development and bench-level deployment systems from Phase I.

Personnel. The largest cost is PI effort. The PI will lead algorithm implementation, dataset preparation, validation design, result interpretation, customer discovery, and Phase II planning. The proposed PI effort exceeds the NSF minimum Phase I commitment and is necessary because the technical architecture and domain assumptions remain founder-led at this stage.

Technical implementation and validation support. Funds are allocated for targeted software engineering, data preparation, and external technical review. The external reviewer or consultant is essential to mitigate solo-founder risk and to ensure that the reported results are not merely founder-confirmed outputs. Consultant work will focus on reproducibility, spectral calculations, SNR improvement, detection-latency comparison, and claim control.

Computer services and software. Funds are allocated for cloud compute, local secure storage, dataset handling, version-control infrastructure, analysis notebooks, and simulation runs. These costs support the 65,536-point FFT sweeps, synthetic artifact-injection experiments, and repeatable analysis across benchmark data. These are software and compute costs, not hardware-infrastructure costs.

Travel and customer discovery. Limited travel is budgeted for customer discovery, industrial partner meetings, and regional validation discussions. The purpose is not broad sales activity. The purpose is to

define Phase II pilot requirements, collect operational constraints, and document customer willingness to evaluate a software-first diagnostic tool.

Materials and supplies. Modest funds are allocated for data-storage media, secure backup, development peripherals, and documentation materials required for reproducible software research. No custom test stand or deployment hardware is proposed for Phase I.

Indirect costs and small business fee. The final Research.gov budget will include allowable indirect costs and the standard small business fee in accordance with NSF and federal cost principles. The final category-level budget should be reviewed by an SBIR-experienced accountant or SBDC advisor before submission. The draft budget contains no standalone contingency line.

12. Data Management and Sharing Plan

This Phase I project will use public benchmark datasets, generated analysis outputs, synthetic artifact-injection records, and limited customer-discovery documentation. Public benchmark references will be documented so that reviewers can identify the original data sources. Raw public datasets will not be republished unless licensing permits. Derived analysis outputs will be stored in versioned folders with clear metadata.

The project will retain raw input references, preprocessing scripts, analysis notebooks, configuration files, feature tables, spectral output summaries, and validation reports. Each experiment will include metadata for dataset source, signal channel, sampling rate, preprocessing steps, transform parameters, window settings, artifact-injection parameters where applicable, and baseline comparator used. This structure is intended to make the Phase I result auditable.

Proprietary SINE~WaiV source code, scheduler logic, scoring heuristics, compiled binaries, and commercialization materials will be protected under SBIR data rights and company trade-secret procedures. Summary results, high-level validation statistics, non-proprietary methodology descriptions, and final technical conclusions may be included in NSF reports, advisor materials, and future investor or partner discussions. Any field data obtained later will be de-identified and controlled through written data-use agreements.

13. Post-Phase I and Phase II Pathway

If Phase I is successful, Phase II will move from benchmark and controlled contamination validation into field evaluation. The Phase II technical focus would be shadow-mode deployment on live VFD-driven assets, comparison against maintenance records and available vibration or thermal observations, and development of a repeatable operator-facing diagnostic report. Hardware validation and edge-gateway deployment pathways are Phase II activities pending successful Phase I outcomes.

The Phase II commercial focus would be partner integration. This may include pilot agreements with industrial operators, a limited beta release for reliability service providers, and discussions with VFD or automation platform vendors. The commercial metric is not only technical performance but adoption friction: how difficult it is to obtain data, how much trust operators place in the output, how the tool fits maintenance workflows, and what integration model produces a viable purchasing decision.

A successful Phase I will therefore produce more than plots. It will produce a bounded technical claim, a reproducible validation package, an independent review memorandum, a customer-discovery record, and a Phase II work plan. That package is the correct bridge from founder-led proof of concept to fundable field validation.

14. Facilities, Equipment, and Other Resources

AI BASE3 LLC will conduct the Phase I work using a software-first development environment. Required resources include a secured development workstation, version-controlled source repository, structured data storage, analysis notebooks, and access to public benchmark datasets. The work does not require construction of a custom laboratory apparatus during Phase I. That boundary is intentional. The purpose of Phase I is to decide whether the signal-processing architecture is technically feasible before requesting funds for broader field deployment or bench-level validation.

The PI's primary facility requirement is a controlled software and data-management environment with sufficient compute capacity for repeated FFT sweeps, synthetic artifact-injection experiments, feature extraction, and reproducibility checks. Cloud compute may be used for parallel validation runs, parameter sweeps, and storage redundancy. Local storage will preserve raw references, scripts, configuration files, and derived outputs. All analysis artifacts will be versioned so that independent reviewers can trace results from input data to reported metrics.

External resources will be used selectively. A technical consultant will support independent validation, methods review, and claim discipline. Regional small-business, manufacturing-extension, or commercialization resources may support customer discovery and Phase II pilot planning. Any future plant partner will be treated as a data-access and operational-context partner, not as a required Phase I hardware host. This resource posture keeps the project aligned with the Phase I objective: software validation of motor-current signal intelligence under VFD artifact constraints.

15. Claim Control and Reviewer Risk Matrix

The proposal also uses explicit decision gates. Gate 1, at the end of Month 3, determines whether the canonical build reproduces baseline results without hidden parameter changes. If Gate 1 fails, the project stops broad validation and repairs reproducibility. Gate 2, at the end of Month 5, determines whether the temporal scoring layer behaves consistently on the NASA IMS proxy benchmark. If Gate 2 fails, the scoring layer is revised before current-domain claims are attempted. Gate 3, at the end of Month 8, determines whether Paderborn current-channel data supports repeatable fault-feature extraction. If Gate 3 fails, the Phase II plan shifts away from field deployment and toward additional current-domain research. Gate 4, at the end of Month 10, determines whether synthetic VFD artifact injection supports the targeted suppression claims. If Gate 4 fails, all commercial claims about VFD artifact suppression are removed or narrowed. Gate 5, at the end of Month 11, determines whether the independent reviewer agrees that the supported claims are reproducible and properly bounded.

These gates are deliberately strict because the proposal is high-risk, high-impact engineering research. The strongest fundable posture is not to promise certainty. The strongest posture is to show NSF that the company knows exactly what would prove feasibility, what would disprove feasibility, and what would trigger a narrowed Phase II path.

This draft intentionally controls claim strength. Reviewer confidence is increased when every claim is either supported, testable, or clearly identified as a Phase I hypothesis. The following claim-control posture governs the proposal.

Claim 1: SINE~WaiV is software-only in Phase I. This is a settled scope boundary. The proposal does not request funds to build a custom test stand, deploy custom sensor hardware, or manufacture edge devices. It requests funds to build, test, validate, and independently review the signal-processing software layer.

Claim 2: NASA IMS supports timing analysis only. The proposal does not present NASA IMS as current-domain proof. NASA IMS is valuable because it provides a known run-to-failure sequence for degradation scoring. The current-domain proof must come from Paderborn current-channel analysis and controlled VFD artifact injection tests.

Claim 3: The FFT architecture is enabling, not magic. Mixed-radix computation is known. The contribution is the constrained application of a 1,728-point, 28-divisor sub-window schedule to VFD carrier-artifact avoidance in MCSA. That is the proper reviewer-facing novelty statement.

Claim 4: The 170-185 Hz and 258-260 Hz kill zones are Phase I test bands. They are not presented as universal bands across every drive. They are the artifact regions selected for the feasibility test. Phase I results will determine whether the method generalizes or requires per-drive calibration.

Claim 5: The PI's nontraditional background is a strength but not a substitute for validation. The founder's field experience explains why the problem matters and why the solution must be deployable in real plants. Independent technical review is included because reviewer confidence also requires external scrutiny.

Claim 6: Commercialization is credible only if the integration burden is low. The proposal therefore avoids claiming that plants will immediately replace installed reliability platforms. The first commercial posture is an overlay, licensing, or feature-extraction engine that can augment existing maintenance workflows.

These claim controls directly answer the most likely reviewer objections: implementation novelty, dataset-domain mismatch, solo-founder execution risk, current-domain evidence, and commercialization friction.

16. Administrative Readiness and Research.gov Assembly Notes

The full Research.gov submission will require more than this narrative draft. This document is intended to serve as the canonical technical and commercialization text from which the final uploaded components can be created. Before submission, the company should confirm active SAM status, UEI accuracy, legal business name, tax identification details, bank information if required, Project Pitch invitation validity, and PI primary-employment compliance at the time of award. NSF requires an official Project Pitch invitation for Phase I proposals, and invitations are valid only for the applicable submission windows.

The final Research.gov package should be assembled by separating this single-file draft into the required components: Project Summary, Project Description, References Cited, Biographical Sketch or personnel information, Budget and Budget Justification, Facilities/Equipment/Other Resources, Data Management and Sharing Plan, Company/Commercialization materials as required, and the SBIR/STTR questionnaire and certifications. The budget entered in Research.gov must match the budget justification. The final budget should be reviewed by an SBIR-experienced accountant, SBDC advisor, or other qualified reviewer before upload.

The final technical edit should also check page limits, fonts, margins, citations, file naming, proprietary legends, and consistency between all forms. A reviewer should not see conflicting names, legacy project labels, unsupported customer references, or broader Phase II activities described as Phase I deliverables. The clean submission story is: SINE~WaiV is a software-only current-domain signal intelligence engine; Phase I validates whether its mixed-radix sub-window scheduler improves fault observability under VFD artifact conditions; independent review controls the solo-founder risk; commercialization proceeds through low-friction software overlay, service-provider, and OEM licensing channels.

17. Customer Discovery Evidence Plan

A major reviewer risk for early SBIR proposals is that the commercialization section sounds plausible but lacks evidence from the market. Phase I therefore includes a customer-discovery evidence plan. The PI will interview maintenance managers, plant electricians, automation engineers, reliability consultants, and industrial decision makers who understand VFD-driven assets. Each interview will be documented with the same core questions: Which motors create the highest downtime risk? What condition-monitoring tools are currently used? What data can be accessed from the VFD or motor-control center? What prevents broader monitoring coverage? What proof would be required before a paid pilot? What integration path would be least disruptive? What price structure would be acceptable if the tool reduced sensor installation or improved warning time?

The output will be a structured evidence log, not a marketing anecdote list. The log will categorize each contact by customer type, asset class, monitoring method, buying authority, integration barrier, and pilot-readiness level. This directly supports the Phase II plan because it converts general market interest into a field-trial funnel. A strong Phase I commercialization result would identify at least three credible pilot candidates, at least one reliability-service channel candidate, and at least one OEM or platform-contact pathway. A weak result would still be useful if it identifies the reasons the market is not ready, such as data-access limitations, cybersecurity approval barriers, uncertainty about VFD waveform availability, or preference for established vibration systems.

This evidence plan keeps the commercialization claim grounded. The proposal does not assume that industrial operators will buy an algorithm because the math is interesting. Operators buy reduced downtime, fewer emergency calls, lower inspection burden, safer planning windows, and easier maintenance decisions. Phase I will test whether SINE~WaiV can be translated into those buying reasons. The customer-discovery work is therefore a parallel risk-reduction activity alongside technical validation.

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19. Submission Cleanup Checklist

Before submission, the founder should verify that the Research.gov package matches this narrative and that the following items are complete: active SAM registration, current UEI, confirmed company legal name, Project Pitch invitation validity, accurate PI employment commitment, final Research.gov budget by NSF category, current and pending support if required, biosketch or personnel documents, facilities/equipment document, company commercialization history if applicable, SBIR/STTR certification forms, proprietary legend decisions, and final PDF formatting under the applicable NSF instructions. This draft applies the current scope-control decisions: excluded legacy dataset language, excluded customer-specific legacy references, no external doctrine branding, no Phase I bench-level hardware validation, and SINE~WaiV positioned as a software-only motor-current signal intelligence effort.